DEENBANDHU CHHOTU RAM UNIVERSITY OF SCIENCE AND TECHNOLOGY, MURTHAL ,SONIPAT

Summer Training Report on Artificial Intelligence Technologies

Submitted in fulfilment of the requirements for completion of 4th semester of

BACHELOR OF TECHNOLOGY (CSE)

Submitted by
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Acknowledgement

I would like to express my profound gratitude to my professors for providing me theopportunity to undergo a 6 week industrial training

I would like to express my special thanks to my mentor Mrs. Nidhi Chawla at THE INSTITUTE NIELT DELHI for her time and efforts that she had put in order to bring the best out of me. Your useful advice and suggestions were really helpful to me during the project's completion. In this aspect, I am eternally grateful to you.

I would like to acknowledge that the project during me training was entirely done by me andnot by someone else.

Sakshi

(22001001907)

CSE 3rd Year

INDEX

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Ministry of Electronics & Information Technology
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Course Completion Letter

It is certified that

SAKSHI ARYA

has successfully undergone the Bridge Course Training Program on

ARTIFICIAL INTELLIGENCE TECHNOLOGIES

under 'FutureSkills PRIME' programme

held from **27th Jun 2023 to 10th Aug 2023**

Chief Investigator
Artificial Intelligence Co-Lead
Resource Center

Signature Not Verified
Digitally signed by SANJAY KUMAR DHURANDHER
Date: 2023.09.24 11:50:29 IST

Executive Director

Introduction: About Organisation

National Institute of Electronics & Information Technology (NIELIT),(erstwhile DOEACC Society), an Autonomous Scientific Society under the administrative control of Ministry of Electronics & Information Technology (MoE&IT), Government of India, was set up to carry out Human Resource Development and related activities in the area of Information, Electronics & Communications Technology (IECT). NIELIT is engaged both in Formal & Non-Formal Education in the area of IECT besides development of industry oriented quality education and training programmes in the state-of-the-art areas. NIELIT has endeavoured to establish standards to be the country‟s premier institution for Examination and Certification in the field of IECT. It is also one of the National Examination Body, which accredits institutes/organizations for conducting courses in IT in the non formal sector

Training Schedule

Week 1 : Basics of python

Week 2:Introduction to AI

Week3: Introduction to ML

Week 4: Implementation of ML Algo.

Week 5: Project

Week 6 : Project

Introduction to AI

- Enabling computers to mimic human cognitive activity.
- Traditional computers do not have thinking capabilities.
- They rely on hard-coded rules (algorithms) to perform computation.
- The goals of artificial intelligence include learning, reasoning, and perception.
- Artificial intelligence is the broader concept that consists of everything from Good Old-Fashioned AI, all the way to futuristic technologies such as deep learning.

Definition of AI

Artificial Intelligence – the attempt to get computers to do things that if people did them we would say that was intelligent.

Examples:- problem solving , planning , game playing , conversation , Making music and art, learning , teaching ,answering questions, Recommendation, Alexa, Google Assistant, Snapchat AI .

Q. What is the focus of AI research?

1. To replace humans .
2. To emulate humans.
3. To perform certain task better than humans .
4. None are correct.

Answer :-

INTRODUCTION TO AI

The science of making machines that,

- Think like people.
- Act like people.
- Think rationally.
- Act rationally.

Artificial Intelligence is an approach to make a computer , or a robot, or a product to think how smart humans think.

THE EVOLUTION OF AI

- Artificial Intelligence :- Engineering of making intelligent Machine and Programs.
  Duration :- 1950's to 1980's.

- Machine Learning :- Ability to learn without being explicity programmed.
  Duration :- 1980's to 2006's.

- Deep Learning:- learning based on deep nueral network.
  Duration :- 2010's to 2017's

Types of AI

It can be categories in two types :-

- Strong AI
- Week AI

STRONG AI

- Allows a machine to apply knowledge and skills in different contexts.
- It is a theoretical form of AI used to describe a certain mind-set of AI development.
- If researchers are able to develop Strong AI, the machine would require an intelligence equal to humans.
- It would have a self-aware consciousness that has the ability to solve problems, learn, and plan for the future.

WEAK AI

-
- Focusses on a narrow task
- Extremely efficient at just the task
- Non intelligent or unable to perform anything outside the task.
- Weak AI relies on human interference to define the parameters of its learning algorithms and to provide the relevant training data to ensure accuracy.
- Examples of weak AI are virtual assistants (like Siri, Google Assistant), chess playing AI, self driving cars and many more!

TERMINOLOGIES OF AI

Artificial Intelligence (AI)
A field of computer science dedicated to the study of computer software making intelligent decisions, reasoning, and problem solving.

Algorithm
A formula or set of rules for performing a task. In AI, the algorithm tells the machine how to go about finding answers to a question or solutions to a problem.

Autonomous
Autonomy is the ability to act independently of a ruling body. In AI, a machine or vehicle is referred to as autonomous if it doesn't require input from a human operator to function properly.

Machine learning
A field of AI focused on getting machines to act without being programmed to do so. Machines "learn" from patterns they recognize and adjust their behavior accordingly.

Deep learning
A subset of machine learning that uses specialized algorithms to model and understand complex structures and relationships among data and datasets.

Natural language processing (NLP)
The ability of computers to understand, or process natural human languages and derive meaning from them. NLP typically involves machine interpretation of text or speech recognition.

Planning
A branch of AI dealing with planned sequences or strategies to be performed by an AI-powered machine. Things such as actions to take, variable to account for, and duration of performance are accounted for.

Pruning
The use of a search algorithm to cut off undesirable solutions to a problem in an AI system. It reduces the number of decisions that can be made by the AI system.

Turing test
A test developed by Alan Turing that tests the ability of a machine to mimic human behavior. The test involves a human evaluator who undertakes natural language conversations with another human and a machine and rates the conversations.

APPLICATIONS OF ARTIFICIAL INTELLIGENCE

- We find its applications in areas as diverse as
  - Image recognition

- Image search
- Object detection
- Computer vision
- Scene/Video parsing
- Face recognition
- Pose estimation

INTRODUCTION TO PYTHON

Python is a popular programming language .It was created by Guido Rossum , and released in 1991. It is used for web development (Server-side), software development, Mathematics, System Scripting.

Scripting Mode:
Besides the interactive mode, you can write Python code in files with a .py extension and execute them using the command python filename.py. This is known as scripting mode and is the most common way to write Python programs.

Basic Syntax: -

- Python uses indentation (whitespace) to define blocks of code, unlike languages that use braces or other delimiters. This enforces clean and readable code.

- Statements are typically terminated with a newline character, but we can use a semicolon (;) to separate multiple statements on a single line.

Variables and Data Types:

- Variables are used to store data. You can assign values to variables using the assignment operator (=).
- Common data types in Python include integers (int), floating-point numbers (float), strings (str), booleans (bool), and more.
- Python is dynamically typed, which means you don't need to declare a variable's type explicitly. Python infers the type based on the assigned value.

Control Structures:

- Python provides constructs like if, elif, and else for conditional branching.
- Loops can be created using for and while statements.
- Python has a range() function for generating sequences of numbers in loops.

Functions:

- Functions are defined using the def keyword. They allow you to encapsulate blocks of code for reuse.
- Functions can accept parameters and return values using the return statement.

Libraries and Modules:

- Python has a vast standard library that provides pre-built modules and packages for various tasks, making development faster and more accessible.
- You can also install third-party libraries using tools like pip to extend Python's functionality.

Here are some of the most popular libraries and packages used in Python:

1. NumPy: NumPy is a fundamental library for numerical computing in Python. It provides support for multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on them efficiently.

2.Pandas: Pandas is a library for data manipulation and analysis. It offers data structures like Data Frames and Series, which are particularly useful for working with structured data.

3.Matplotlib: Matplotlib is a powerful library for creating static, animated, and interactive visualizations in Python. It is commonly used for data visualization and plotting.

4.Seaborn: Seaborn is built on top of Matplotlib and provides a high-level interface for creating informative and attractive statistical graphics.

5.Scikit-Learn: Scikit-Learn is a machine learning library that offers tools for classification, regression, clustering, dimensionality reduction, and more. It is widely used in machine learning projects and research.

WHAT CAN PYTHON DO?

- Python can be used on a server to create web applications.
- Python can be used alongside software to create workflows.
- Python can connect to database systems. It can also read and modify files.
- Python can be used to handle big data and perform complex mathematics.
- Python can be used for rapid prototyping, or for production-ready software development.

WHY PYTHPON

- way, an object-oriented way or a functional way.
- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc).
- Python has a simple syntax similar to the English language.
- Python has syntax that allows developers to write programs with fewer lines than some other programming languages.
- Python runs on an interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick. Python can be treated in a procedural

INTRODUCTION TO MACHINE LEARNING

Machine Learning (ML) is a subfield of artificial intelligence (AI) that focuses on the development of algorithms and statistical models that enable computers to learn and make predictions or decisions without being explicitly programmed for a specific task. In essence, it's about creating systems that can learn from data and improve their performance over time. Here's a comprehensive introduction to the key concepts in machine learning:

WHY IS MACHINE LEARNING IMPORTANT?

- Data is the lifeblood of all business.
- Data-driven decisions increasingly make the difference between keeping up with competition or falling further behind
- Machine learning can be the key to unlocking the value of corporate and customer data and enacting decisions that keep a company ahead of the competition.

HOW MACHINE LEARNING WORKS

Data Collection:-
Data collection is the first step of the machine learning life cycle. The goal of this step is to identify and obtain all data-related problems.
In this step, we need to identify the different data sources, as data can be collected from various sources such as files, database, internet, or mobile devices. It is one of the most important steps of the life cycle. The quantity and quality of the collected data will determine the efficiency of the output. The more will be the data, the more accurate will be the prediction.

o Identify various data sources
o Collect data
o Integrate the data obtained from different sources

By performing the above task, we get a coherent set of data, also called as a dataset. It will be used in further steps.

Data preparation:-
After collecting the data, we need to prepare it for further steps. Data preparation is a step where we put our data into a suitable place and prepare it to use in our machine learning training.

In this step, first, we put all data together, and then randomize the ordering of data.
This step can be further divided into two processes:

- Data exploration:

It is used to understand the nature of data that we have to work with. We need to understand the characteristics, format, and quality of data.
A better understanding of data leads to an effective outcome. In this, we find Correlations, general trends, and outliers.

- Data pre-processing:

Now the next step is preprocessing of data for its analysis.

Data Wrangling:-
Data wrangling is the process of cleaning and converting raw data into a useable format. It is the process of cleaning the data, selecting the variable to use, and transforming the data in a proper format to make it more suitable for analysis in the next step. It is one of the most important steps of the complete process. Cleaning of data is required to address the quality issues.
It is not necessary that data we have collected is always of our use as some of the data may not be useful. In real-world applications, collected data may have various issues, including:

- Missing Values
- Duplicate data
- Invalid data
- Noise

So, we use various filtering techniques to clean the data.
It is mandatory to detect and remove the above issues because it can negatively affect the quality of the outcome.

Data Analysis:-
Now the cleaned and prepared data is passed on to the analysis step. This step involves:

- Selection of analytical techniques
- Building models
- Review the result

The aim of this step is to build a machine learning model to analyse the data using various analytical techniques and review the outcome. It starts with the determination of the type of the problems, where we select the machine learning techniques such as Classification, Regression, Cluster analysis, Association, etc. then build the model using prepared data, and evaluate the model.

Model Training:
The selected algorithm is trained on the training data. During training, the model adjusts its internal parameters to minimize the difference between its predictions and the actual target values in the training data. This process continues until the model reaches a satisfactory level of accuracy.

Test Model:-
Once our machine learning model has been trained on a given dataset, then we test the model. In this step, we check for the accuracy of our model by providing a test dataset to it.
Testing the model determines the percentage accuracy of the model as per the requirement of project or problem.

Deployment:-
The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system. If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system. But before deploying the project, we will check whether it is improving its performance using available data or not. The deployment phase is similar to making the final report for a project.

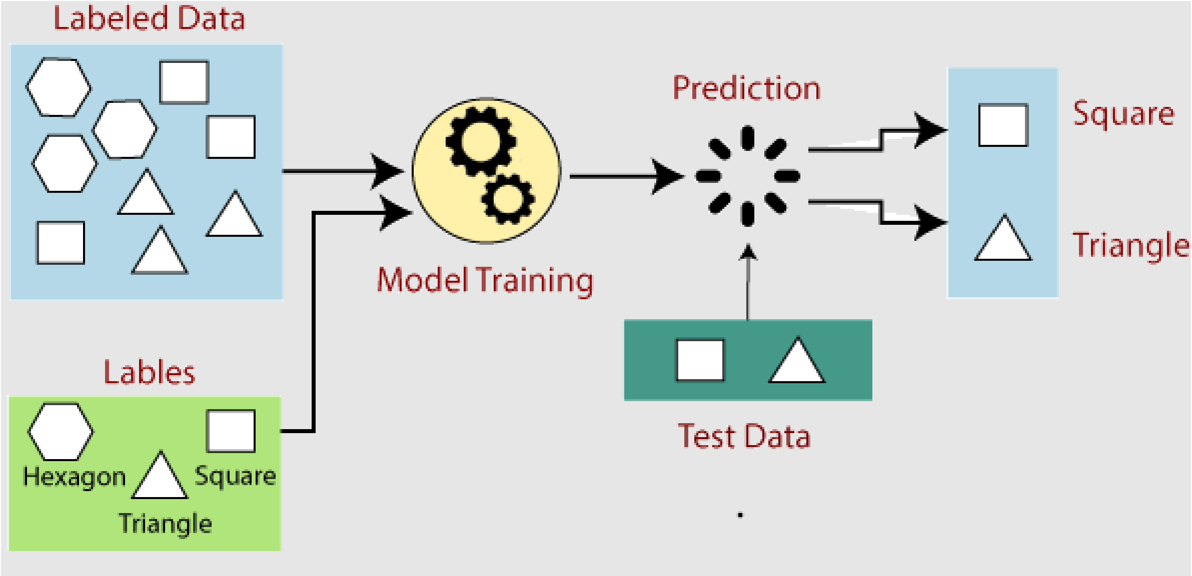
Hyperparameter Tuning:
Hyperparameters are settings that control the learning process of the algorithm (e.g., learning rate, depth of a decision tree). Tuning these hyperparameters is essential to optimize the model's performance.

Model Evaluation:
The model's performance is assessed using the testing/validation set. Common evaluation metrics depend on the problem type and can include accuracy, precision, recall, mean squared error, and more.

Model Deployment:
Once the model performs well on the testing/validation set, it can be deployed in a real-world application. This involves integrating the model into a software system or platform where it can make predictions or decisions based on new, unseen data.

Continuous Learning:
Machine learning models require maintenance and continuous monitoring. New data is collected over time, and the model may need to be retrained periodically to stay accurate and up-to-date.

Ethical Considerations:
Throughout the machine learning process, ethical considerations such as fairness, bias, and privacy should be addressed. It's important to ensure that the model's predictions do not discriminate against certain groups or violate privacy norms.
Machine learning is a dynamic field that continually evolves with new algorithms, techniques, and applications. It has a wide range of real-world applications, including natural language processing, image recognition, recommendation systems, autonomous vehicles, and more. The success of a machine learning project depends on the quality of data, the choice of algorithm, and the careful tuning and evaluation of the model.



TOP MACHINE LEARNING USE CASES?

THREE COMPONENTS OF MACHINE LEARNING

Data:-

- Want to detect spam? Get samples of spam messages.
- Want to forecast stocks? Find the price history.
- It's extremely tough to collect a good collection of data (usually called a dataset).

Features:-

- Also known as parameters or variables.
- E.g. car mileage, user's gender, stock price, word frequency in the text.
- In other words, these are the factors for a machine to look at.

Algorithms:-

- receive and analyze input data to predict output values.
- As new data is fed to these **algorithms**, they learn and optimize their operations to improve performance, developing '**intelligence**' over time.

TYPES OF MACHINE LEARNING

Machine learning can be categorized into several types based on the nature of the learning process and the kind of data used. The main types of machine learning are:

1.Supervised Learning:

- In supervised learning, the algorithm is trained on a labeled dataset, where each input is associated with a corresponding target or output. The goal is to learn a mapping from inputs to outputs.
- Common supervised learning tasks include classification (e.g., spam email detection, image recognition) and regression (e.g., predicting house prices).

2.Unsupervised Learning:

- Unsupervised learning deals with unlabeled data, where the algorithm tries to find patterns, structures, or relationships within the data without the guidance of labeled outputs.
- Common unsupervised learning tasks include clustering (e.g., grouping similar customer profiles) and dimensionality reduction (e.g., reducing the number of features in a dataset).

3.Reinforcement Learning:

- Reinforcement learning involves an agent that learns to make sequential decisions by interacting with an environment. The agent receives rewards or penalties based on its actions and learns to maximize its cumulative rewards over time.
- Applications of reinforcement learning include game playing (e.g., AlphaGo), robotics, and autonomous vehicle control.

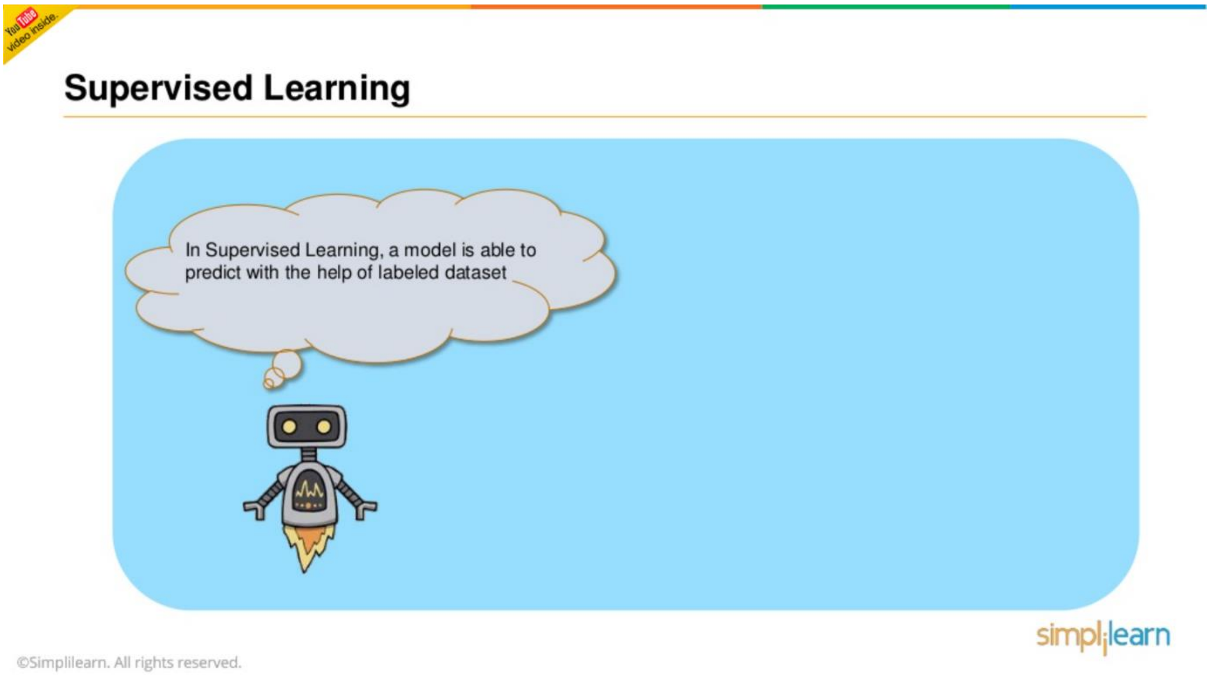
SUPERVISED LEARNING

The machine has a "supervisor" or a "teacher" who gives the machine all the answers, like whether it's a cat in the picture or a dog. The teacher has already divided (labeled) the data into cats and dogs, and the machine is using these examples to learn. One by one.

HOW TO WORK SUPERVISED LEARNING

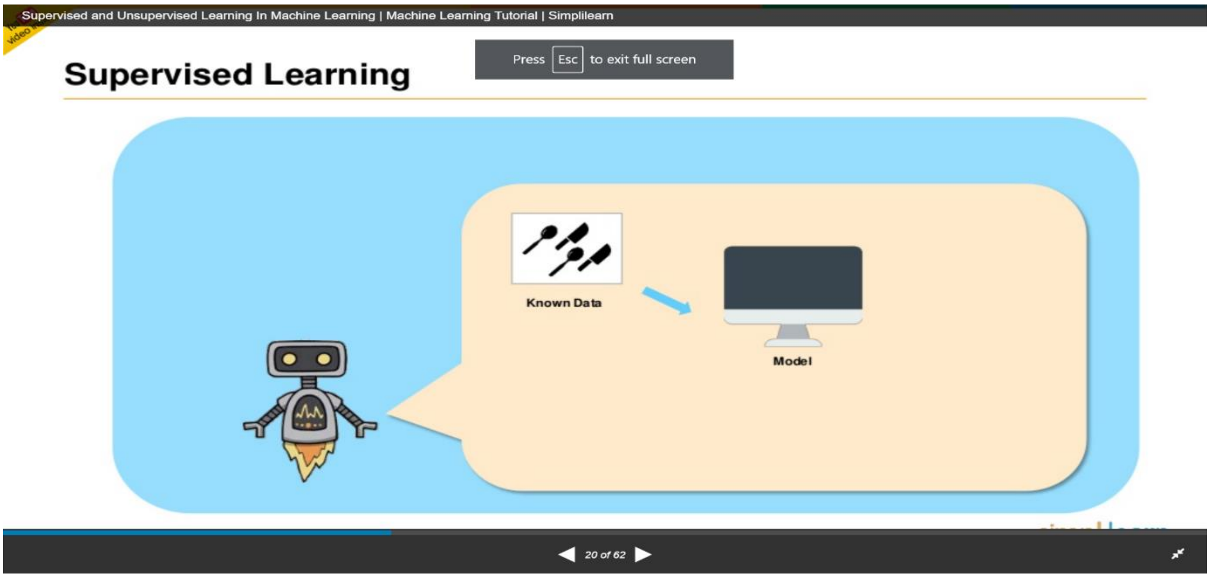
Step 1:- Data collection:-

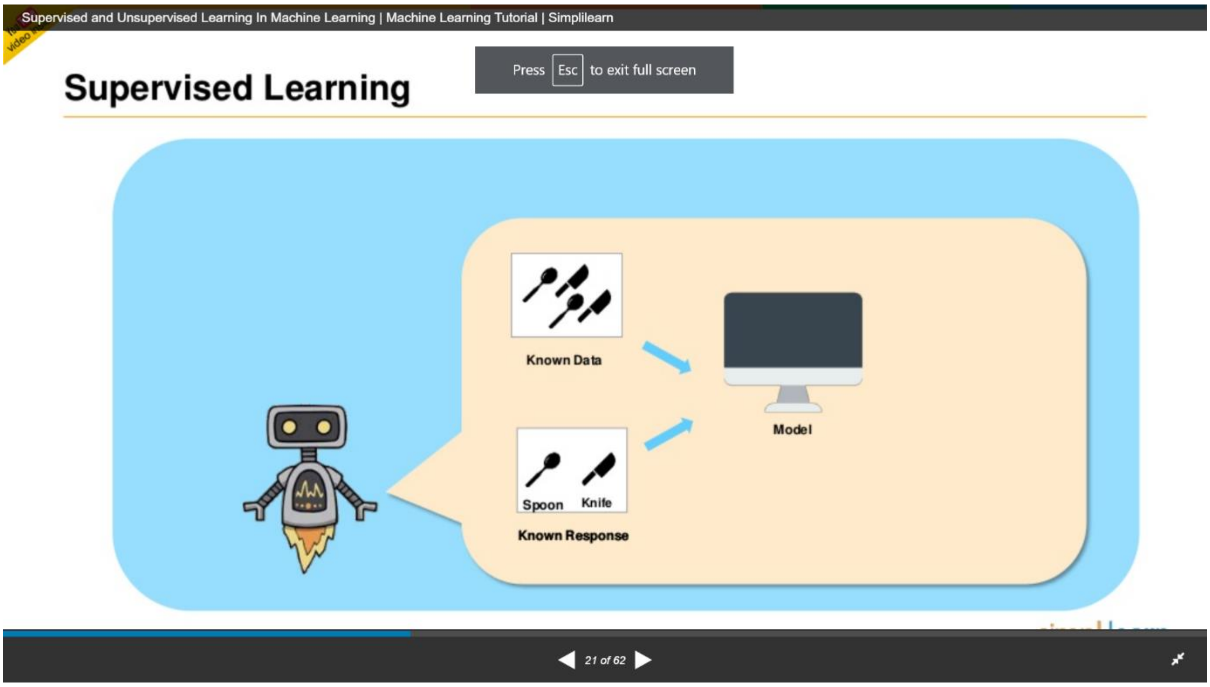
Gather a dataset that consists of pairs of input examples and their corresponding correct output labels. This dataset is called the "training dataset."



Data Preprocessing:

- Clean and preprocess the data to make it suitable for training.
- This includes handling missing values, scaling features, and encoding categorical variables.





Data Splitting:

- Divide the dataset into two subsets: the "training set" and the "testing set" (or validation set).
- The training set is used to train the model, while the testing set is used to evaluate its performance.

REGRESSION IN MACHINE LEARNING: -

Regression in machine learning is a supervised learning technique used to model the relationship between a dependent variable (also known as the target or response variable) and one or more independent variables (predictors or features) by fitting a mathematical function or model to the data. The primary goal of regression is to predict a continuous numeric output based on the input features. Here are some key aspects of regression in machine learning:

Types of Regression:

- **Linear Regression**: Linear regression models the relationship between the dependent variable and one or more independent variables as a linear equation. It is one of the simplest and most widely used regression techniques.

- **Logistic Regression**: Despite its name, logistic regression is used for binary classification problems rather than regression. It models the probability that an input belongs to a specific class.

Objective:-
The objective of regression is to find the best-fitting model that minimizes the difference between the predicted values (output of the model) and the actual observed values in the training data. Common loss functions for regression include Mean Squared Error (MSE) and Mean Absolute Error (MAE).

Model Evaluation:- Regression models are evaluated using metrics that quantify the accuracy and goodness of fit of the predictions. Common evaluation metrics include R-squared (coefficient of determination), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

Multiple Features:
In multiple linear regression, there can be multiple independent variables, allowing you to model complex relationships between the target variable and the predictors.
Overfitting and Underfitting: Like other machine learning models, regression models can suffer from overfitting (fitting the training data too closely and failing to generalize) or underfitting (being too simple to capture the underlying patterns). Regularization techniques like Ridge and Lasso regression can help mitigate overfitting.

Feature Selection:-
Feature selection is the process of choosing the most relevant independent variables for a regression model. It can improve model performance and reduce overfitting.
Applications: Regression is widely used in various fields, including finance (predicting stock prices), economics (modelling demand and supply), healthcare (predicting patient outcomes), and natural sciences (fitting experimental data to mathematical models).

Assumptions:-
Linear regression, in particular, relies on certain assumptions, such as linearity, independence of errors, homoscedasticity (constant variance of errors), and normality of errors. Violations of these assumptions can affect the model's reliability.
Interpretability: Linear regression models are highly interpretable because they provide coefficients that indicate the strength and direction of the relationship between each predictor and the target variable.
Regression is a versatile and fundamental technique in machine learning and statistics, and it

serves as a building block for more complex models and predictive analytics tasks. It is especially useful when you want to make predictions based on continuous numerical data and understand the quantitative relationships between variables.

FEATURE SELECTION TECHIQUE:-

Feature selection is a crucial step in the machine learning pipeline, as it involves choosing a subset of relevant features (independent variables) from the original set of available features. The goal is to improve model performance, reduce overfitting, and speed up training while maintaining or even enhancing the interpretability of the model. There are several feature selection techniques, broadly categorized into three groups: filter methods, wrapper methods, and embedded methods.

Filter Methods:

Filter methods evaluate the relevance of each feature independently of the machine learning algorithm used. They are based on statistical or information-theoretic metrics. Some common filter methods include:

Filter methods are computationally efficient and can be applied before model training.

Wrapper Methods:-

Wrapper methods evaluate feature subsets by training and testing machine learning models with different combinations of features. They directly use the performance of the model as the criterion for feature selection.

Embedded Methods:

Embedded methods perform feature selection as an integral part of the model training process. They use the learning algorithm's built-in feature selection techniques to identify the most relevant features.

Embedded methods are efficient because they perform feature selection while training the model itself.

DECISION TREE:_

A decision tree is a popular machine learning algorithm used for both classification and regression tasks. It is a tree-like structure where internal nodes represent feature tests, branches represent decision rules, and leaf nodes represent class labels or regression values. Decision trees are known for their simplicity, interpretability, and ability to handle both categorical and numeric data. Here are key characteristics and concepts related to decision trees:

Tree Structure: A decision tree is a hierarchical structure consisting of nodes. The top node is called the "root," and the final nodes are called "leaves." Internal nodes represent features or attributes, and branches represent possible values or outcomes based on feature tests.

Feature Selection: At each internal node, the decision tree algorithm selects the feature that provides the best split or separation of the data into distinct classes or regression values. This is typically done using criteria like Gini impurity (for classification) or mean squared error (for regression).

Splitting Criteria:

For classification: Common criteria include Gini impurity, entropy, and misclassification

rate. These criteria measure the impurity or disorder of class labels at each node.
For regression: The mean squared error (MSE) and mean absolute error (MAE) are often used to measure the variance of regression values at each node.
Splitting Process: The decision tree algorithm recursively partitions the dataset based on the selected feature and its possible values, creating child nodes for each outcome. This process continues until a stopping condition is met, such as reaching a maximum tree depth or having a minimum number of samples in a leaf.

Pruning: Decision trees can be prone to overfitting, meaning they capture noise in the data and do not generalize well to unseen data. Pruning involves reducing the size of the tree by removing branches that do not contribute significantly to improving predictive accuracy. This helps mitigate overfitting.

Prediction: To make a prediction, you start at the root node and follow the path down the tree, testing each feature along the way, until you reach a leaf node. The class label (for classification) or regression value (for regression) associated with the leaf node is the predicted output.

Interpretability: One of the key advantages of decision trees is their interpretability. You can easily visualize a decision tree and understand the sequence of decisions it makes to arrive at a prediction.

Ensemble Methods: Decision trees are often used as building blocks for ensemble methods like Random Forests and Gradient Boosting. These methods combine multiple decision trees to improve predictive accuracy and robustness.

Handling Missing Data: Decision trees can handle missing values by considering multiple paths when a feature has missing data. This makes them robust to incomplete datasets.

Applications: Decision trees are used in various domains, including finance (credit scoring), healthcare (disease diagnosis), marketing (customer segmentation), and natural language processing (text classification).

RANDOM FOREST CLASSIFIER :-

A Random Forest Classifier is an ensemble learning method based on decision trees. It is a popular and powerful machine learning algorithm that combines the predictions of multiple decision trees to make more accurate and robust classifications. Random Forests are used for both classification and regression tasks and have several advantages, including high accuracy, resistance to overfitting, and the ability to handle large datasets with many features. Here are key features and concepts related to Random Forest Classifier:

Ensemble Learning: Random Forest is an ensemble learning technique, which means it combines the predictions of multiple models to improve overall performance. In the case of Random Forest, the base models are decision trees.

Random Subsampling (Bootstrap Aggregating or Bagging): Random Forest uses a technique called bootstrap aggregating, or bagging, to create multiple subsets (bootstrapped samples) of the training data. Each subset is used to train a separate decision tree.

Feature Randomness: In addition to using bootstrapped samples, Random Forest introduces feature randomness. At each node of each decision tree, only a random subset of features is considered for splitting. This decorrelates the trees and reduces the risk of overfitting.

Voting: For classification tasks, each decision tree in the Random Forest independently predicts the class label of a data point. The final prediction is determined by a majority vote (mode) among the individual tree predictions.

Prediction Confidence: Random Forests can provide a measure of prediction confidence or probability estimates. For classification, they can output class probabilities based on the fraction of trees voting for each class.

Parallelization: Random Forest training can be parallelized easily, making it suitable for multi-core processors and distributed computing environments.

Hyperparameters: Random Forest has several hyperparameters that can be tuned to optimize model performance, including the number of trees in the forest, the maximum depth of the trees, the minimum number of samples required to split a node, and the number of features considered at each split.

Out-of-Bag (OOB) Error: Random Forests can estimate their own generalization error using out-of-bag samples. These are data points not included in the bootstrap sample for each tree. The OOB error can serve as a useful validation metric during training.

Feature Importance: Random Forests can measure the importance of each feature in making predictions. This information can be helpful in feature selection and understanding the impact of different features on the model's performance.

Applications: Random Forests are widely used in various domains, including classification tasks such as spam detection, image recognition, and medical diagnosis. They are also employed for regression tasks like predicting house prices or stock market trends.

K-Means Clustering: -

K-means clustering is a widely used unsupervised machine learning algorithm that groups similar data points into clusters based on their feature similarities. It is a partitioning algorithm that aims to divide a dataset into K clusters, where each data point belongs to the cluster with the nearest mean (centroid). Here's how K-means clustering works:

Initialization: Choose the number of clusters (K) you want to create. Initialize K cluster centroids randomly within the data's feature space. These initial centroids are usually selected from the data points themselves or using other methods, such as k-means++.

Assignment Step (Expectation): For each data point, calculate its distance to each of the K centroids using a distance metric such as Euclidean distance. Assign the data point to the cluster associated with the nearest centroid.

Update Step (Maximization): After all data points have been assigned to clusters, calculate new centroids for each cluster by taking the mean of all the data points in that cluster. These new centroids represent the center of each cluster.

Iteration: Repeat the assignment and update steps iteratively until one of the stopping criteria is met:

The centroids no longer change significantly between iterations.
A maximum number of iterations is reached.
The assigned data points no longer change clusters significantly.
Final Result: The final result is a set of K clusters, each represented by its centroid. Data points within the same cluster are similar to each other in terms of their feature values.

Key characteristics and considerations of K-means clustering:

Sensitivity to Initial Centroids: The choice of initial centroids can affect the final clustering result. Different initializations can lead to different clustering’s. K-means++ is a commonly used initialization technique to mitigate this issue.

Number of Clusters (K): Determining the appropriate number of clusters (K) is often a challenge. Various methods, such as the elbow method and silhouette analysis, can help you select an optimal K value based on cluster quality metrics.

Scalability: K-means can scale to large datasets, but its performance may degrade with increasing data dimensionality. Preprocessing, such as dimensionality reduction, can be beneficial in such cases.

Non-convex Clusters: K-means assumes that clusters are spherical and equally sized, which may not hold true for all datasets. For complex-shaped clusters, other clustering algorithms like DBSCAN or hierarchical clustering might be more suitable.

Robust to Noise: K-means can be robust to noise because outliers tend to be assigned to the nearest cluster centroid. However, it may struggle with varying cluster densities.

Interpretability: K-means provides cluster centroids that are interpretable and can be useful for understanding the characteristics of each cluster.

Convergence: K-means is guaranteed to converge, but it may converge to a local minimum, so multiple initializations are often used to mitigate this risk.

K-means clustering is widely used in various fields, including image segmentation, customer segmentation, document clustering, and anomaly detection, among others. It is a powerful and efficient technique for discovering patterns and structures in data without the need for labeled target values.

SVM
Support Vector Machine (SVM) is a popular supervised machine learning algorithm used for classification and regression tasks. It is particularly effective for binary classification problems but can be extended to handle multi-class classification as well. SVMs are known for their ability to find optimal hyperplanes that best separate data points belonging to

different classes. Here are the key concepts and characteristics of SVM:

Hyperplane: In binary classification, an SVM seeks to find the optimal hyperplane that best separates two classes in feature space. This hyperplane is chosen such that it maximizes the margin, which is the distance between the hyperplane and the nearest data points from each class. These nearest data points are known as support vectors.

Kernel Trick: SVMs can handle linearly inseparable data by using a kernel function that maps the input data into a higher-dimensional space where it becomes linearly separable. Common kernel functions include linear, polynomial, radial basis function (RBF), and sigmoid kernels.

Margin: The margin is a key concept in SVMs. It represents the distance between the hyperplane and the closest data points (support vectors) from each class. SVM aims to maximize this margin because a larger margin typically results in better generalization to unseen data and improved robustness.

Support Vectors: Support vectors are data points that lie closest to the separating hyperplane and play a crucial role in determining the hyperplane's position. They are the data points that have non-zero coefficients in the SVM's decision function.

Soft Margin: In real-world scenarios, data is often not perfectly separable. SVMs can handle this by introducing a soft margin, allowing for some data points to be on the wrong side of the margin or even the wrong side of the hyperplane. This is controlled by a regularization parameter, often denoted as 'C,' which balances the trade-off between maximizing the margin and minimizing classification errors.

Dual Problem: The optimization problem of finding the hyperplane is often transformed into its dual form, which simplifies the calculations and allows for efficient training, especially when using kernel functions.

Categorical and Multiclass Classification: SVMs can be extended to handle multi-class classification problems using techniques like one-vs-one or one-vs-all, where multiple binary classifiers are trained and combined to make multi-class predictions.

Regression: SVMs can also be used for regression tasks, known as Support Vector Regression (SVR). SVR aims to fit a hyperplane that best captures the relationship between the input features and the target variable.

Outliers: SVMs are robust to outliers because they focus on the support vectors, which are typically not influenced by extreme values.

Interpretability: SVMs provide relatively interpretable models, as the separating hyperplane can be visualized and understood.

Scalability: While SVMs can be powerful for small to moderately sized datasets, they may not be as suitable for large-scale datasets due to their computational complexity.

K-Fold: -
In machine learning, k-fold cross-validation is a commonly used technique for assessing the

performance of a predictive model and for estimating how well the model will generalize to new, unseen data. It is a resampling method that helps in partitioning a dataset into training and testing sets multiple times. Here's how k-fold cross-validation works:

Data Splitting: The original dataset is divided into k roughly equal-sized subsets or "folds." Each fold contains approximately 1/k of the total data.

Iterative Process: The cross-validation process is performed k times. During each iteration, one of the k folds is held out as the test set, and the remaining k-1 folds are used as the training set.

Model Training and Evaluation: For each iteration, a machine learning model is trained on the training set and evaluated on the held-out test set. Performance metrics, such as accuracy, precision, recall, F1-score, or mean squared error, are computed based on the model's predictions on the test set.

Performance Aggregation: The performance metrics obtained in each of the k iterations are typically averaged to provide an overall estimate of the model's performance. This aggregated performance metric, often referred to as cross-validation score, provides a more robust assessment of the model's generalization performance compared to a single train-test split.

Key benefits and considerations of k-fold cross-validation:

Bias and Variance: K-fold cross-validation helps in estimating both bias (underfitting) and variance (overfitting) of a model. If a model performs well on average but has high variance across folds, it might be overfitting the data.

Utilizes Data Efficiently: It makes efficient use of the available data because each data point is used for both training and testing across multiple iterations.

Model Selection: K-fold cross-validation is commonly used for hyperparameter tuning and model selection. You can assess how different hyperparameter settings affect the model's performance.

Robustness: It provides a more reliable estimate of a model's performance compared to a single train-test split, especially when the dataset is small or when data distribution is not uniform.

Randomness: The choice of which data points go into each fold can affect the results. To mitigate this, it's common to use stratified k-fold cross-validation, which preserves the class distribution in each fold.

Computational Cost: Performing k iterations of training and evaluation can be computationally expensive, especially for complex models or large datasets. However, it's a worthwhile trade-off for more accurate performance estimates.

Variations: There are variations of k-fold cross-validation, such as leave-one-out cross-validation (k equals the number of data points), and repeated k-fold cross-validation (repeated k times with different random splits).

K-fold cross-validation is a valuable technique in machine learning for model assessment, selection, and ensuring that the model's performance estimates are robust and reliable. It helps in gaining confidence in how well a model is likely to perform on unseen data.

Project

Case Study 1: Water Quality prediction Using Machine learning Algorithm.

Problem Statement: Safe drinking water is essential to a healthy life . It is a fundamental human right.Healthy drinking water is a vital as a healthy & development issues at a national regional and local level.

Import the libraries:-

```
import pandas as pd
import numpy as np
import seaborn as sns
import plotly.express as px
import matplotlib.pyplot as plt
%matplotlib inline
from sklearn.preprocessing  import  StandardScaler
```

Upload the file

```
[ ]  from google.colab import files
     files.upload()
```

Choose Files water_potability.csv

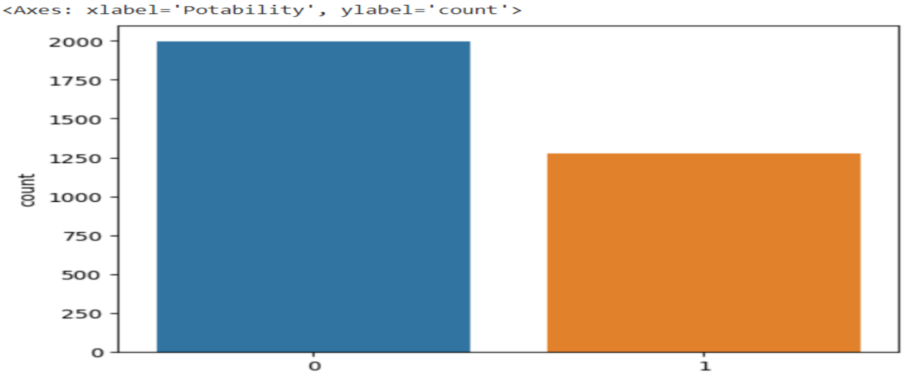
- **water_potability.csv**(text/csv) - 344696 bytes, last modified: 10/3/2023 - 100% done

```
Saving water_potability.csv to water_potability.csv
{'water_potability.csv':
b'ph,Hardness,Solids,Chloramines,Sulfate,Conductivity,Organic_carbon,Triha
```

Data Visualization:-

Seaborn library to create a countplot, a type of bar plot, to visualize the distribution of data in the 'APR Severity of Illness Code' column of the DataFrame 'data'.

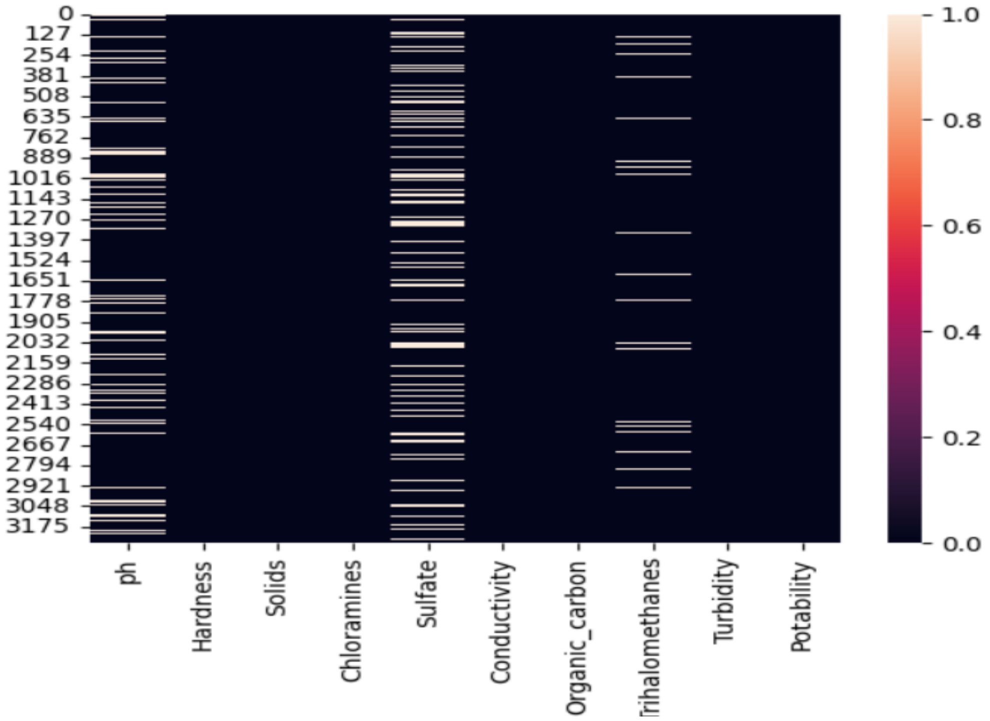
```
sns.countplot(x="Potability" ,data = data)
<Axes: xlabel='Potability', ylabel='count'>
```



creates a heatmap to visualize the correlations between variables in our dataset, making it easier to identify patterns and relationships between different features. The heatmap's color intensity represents the strength and direction of the correlation between pairs of variables, with different colors indicating positive and negative correlations.

```
sns.heatmap(data.isnull())
```

```
<Axes: >
```




```
data.corr()
```

	ph	Hardness	Solids	Chloramines	Sulfate	Conductivity
ph	1.000000	0.082096	-0.089288	-0.034350	0.018203	0.018614
Hardness	0.082096	1.000000	-0.046899	-0.030054	-0.106923	-0.023915
Solids	-0.089288	-0.046899	1.000000	-0.070148	-0.171804	0.013831
Chloramines	-0.034350	-0.030054	-0.070148	1.000000	0.027244	-0.020486
Sulfate	0.018203	-0.106923	-0.171804	0.027244	1.000000	-0.016121
Conductivity	0.018614	-0.023915	0.013831	-0.020486	-0.016121	1.000000
Organic_carbon	0.043503	0.003610	0.010242	-0.012653	0.030831	0.020966
Trihalomethanes	0.003354	-0.013013	-0.009143	0.017084	-0.030274	0.001285
Turbidity	-0.039057	-0.014449	0.019546	0.002363	-0.011188	0.005796
Potability	-0.003556	-0.013837	0.033743	0.023779	-0.023577	-0.008128

Removing null Value Using Mean or Drop method

```
[ ]  data["ph"] =data["ph"].fillna(data["ph"].mean())
     data["Sulfate"] =data["Sulfate"].fillna(data["Sulfate"].mean())
     data["Trihalomethanes"] =data["Trihalomethanes"].fillna(data["Trihalomethanes"].me
```

```
data.isnull().sum()
```

```
ph                 0
Hardness           0
Solids             0
Chloramines        0
Sulfate            0
Conductivity       0
Organic_carbon     0
Trihalomethanes    0
Turbidity          0
Potability         0
dtype: int64
```

Split the dataset for train model

```
:learn.model_selection import train_test_split

,x_test,y_train,y_test = train_test_split(x,y,test_size = 0.2,random_state =38)
```

```
x_train.shape, x_test.shape
```

```
((2620, 9), (656, 9))
```

Logistic Regression

```
[86] from sklearn.model_selection import train_test_split
     X_train,X_test,y_train,y_test = train_test_split(X,y,test_size = .2,random_state =42)
```

```
[87] from sklearn.linear_model import LogisticRegression
     from sklearn.metrics import accuracy_score,confusion_matrix, classification_report
```

```
#Crating A logisticRegression Object
logmodel = LogisticRegression()
```

```
print(confusion_matrix(y_test,y_pred))
```

```
[[   0   20  226    0    2]
 [   0   51 1051    0    8]
 [   0   11 2551    0    5]
 [   0    0    1    0    0]
 [   0    4   68    0    0]]
```

```
[106] print(classification_report(y_test,y_pred))
```

```
              precision    recall  f1-score   support

           0       0.00      0.00      0.00       248
           1       0.59      0.05      0.09      1110
           2       0.65      0.99      0.79      2567
           3       0.00      0.00      0.00         1
           4       0.00      0.00      0.00        72

    accuracy                           0.65      3998
   macro avg       0.25      0.21      0.17      3998
weighted avg       0.58      0.65      0.53      3998
```

```
[105] print(accuracy_score(y_test,y_pred))
```

```
0.6508254127063532
```